

Week 4 Workshop

Download monthly data from ‘Yahoo Finance’ on Microsoft (MSFT) and Apple (AAPL) from Dec 2020 to Dec 2025. Use an annual risk free-rate of 3% per year for the T-bill.

1. Compute the monthly log returns.
2. Compute the Sharpe ratio for Microsoft and Apple.
3. Compute the sample covariance matrix and correlation matrix of the monthly log returns of Microsoft and Apple.
4. Compute the GMV weights for these two assets:

$$\begin{aligned}w_1 &= \frac{\sigma_2^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}, \\w_2 &= 1 - w_1 = \frac{\sigma_1^2 - \sigma_{12}}{\sigma_1^2 + \sigma_2^2 - 2\sigma_{12}}.\end{aligned}$$

5. Explain how the slope coefficient estimate of β_1 in the following regression

$$r_{2,t} = \beta_0 + \beta_1 (r_{2,t} - r_{1,t}) + \epsilon_t$$

is related to the GMV weights computed above.

6. Week 5 Tutorial. Collect the portfolio weights into a $N \times 1$ vector $\mathbf{w} = (w_1, \dots, w_N)'$ and compute the GMV weights for these N assets:

$$\mathbf{w} = \frac{\boldsymbol{\Sigma}^{-1}\mathbf{1}}{\mathbf{1}'\boldsymbol{\Sigma}^{-1}\mathbf{1}}$$

where $\mathbf{1}$ is a $N \times 1$ vector of ones and $\boldsymbol{\Sigma}$ is an $N \times N$ sample covariance matrix.

7. Using the GMV weights, compute the GMV portfolio:

$$r_p = w_1 r_1 + (1 - w_1) r_2.$$

Compute the Sharpe ratio for the GMV portfolio.